

Welcome to the Workshop

From Species Descriptions to Computable Phenotypes: An Introduction to Phenoscript and Ontologies

WHS Methodological Pills



Organizers and Instructors

Organizers

- [Willi Hennig Society](#)
- Amrita Srivathsan (WHS)
- Santiago Catalano (WHS)
- Ambrosio Torres Galvis (WHS)

Instructors

- [Sergei Tarasov](#) (Finnish Museum of Natural History)
- [Giulio Montanaro](#) (Finnish Museum of Natural History)
- [Jennifer Girón](#) (Museum at Texas Tech University)

Post in the Zoom chat the organisms you are working with 🦋

Goals and Schedule

Aims:

- Demonstrate how to use (insect) ontologies and Phenoscript to make traits and descriptions computable
 - We focus on species descriptions
 - We focus on insect/vertebrate traits
 - let Sergei know if you want to configure it for other taxa
 - The approach can be applied to matrices
- Grow and engage the community
- We want your feedback on what can be improved

Material:

<https://github.com/sergeitarasov/whs-phenoscript-2026>

Schedule

- 12:00-12:10 **Introduction to the Workshop** (*Sergei*)
- 12:10-12:40 **Introduction to Insect Ontologies** (*Jennifer*)
- 12:40-13:00 **Phenoscript Workflow** (*Sergei*)
- 13:00-13:40 **Encoding Traits using Phenoscript** (*Giulio*)
- 13:40-13:50 **Querying Phenoscript Descriptions** (*Sergei*)
- 13:50-14:00 **Break**
- 14:00-15:00 **Practical Exercises** (*Giulio, Sergei, Jennifer*)
 - Make at least 5 Phenoscript statements about your favorite species. Check if they are correctly translated into OWL and natural text.

Prerequisites

- **Install required software** (all free):
 - [VS Code](#) — for writing Phenoscript
 - [Docker](#) — to run the necessary software
 - [Protégé](#) — ontology editor
- **Install the Phenoscript extension for VS Code** — follow the [installation instructions](#)
- **Get an [ORCID ID](#)** if you do not have one
- **Prepare trait data** — identify five traits of your favorite species that you would like to code using Phenoscript during the workshop and have them available

Highly recommended before the workshop

- **Join OntoTraits Slack** for discussion (link privately shared)
- **Complete the [Pizza Ontology tutorial](#)** — understand what an ontology is (approximately half a day)
- **Important reading:**
 - Insect Anatomy Ontology — <https://doi.org/10.1093/sysbio/syad025>
 - Phenoscript and semantic descriptions — <https://bdj.pensoft.net/article/121562/>

If you did the Pizza Ontology tutorial, post "I like pizza" in the Zoom chat 🍕

Let's start